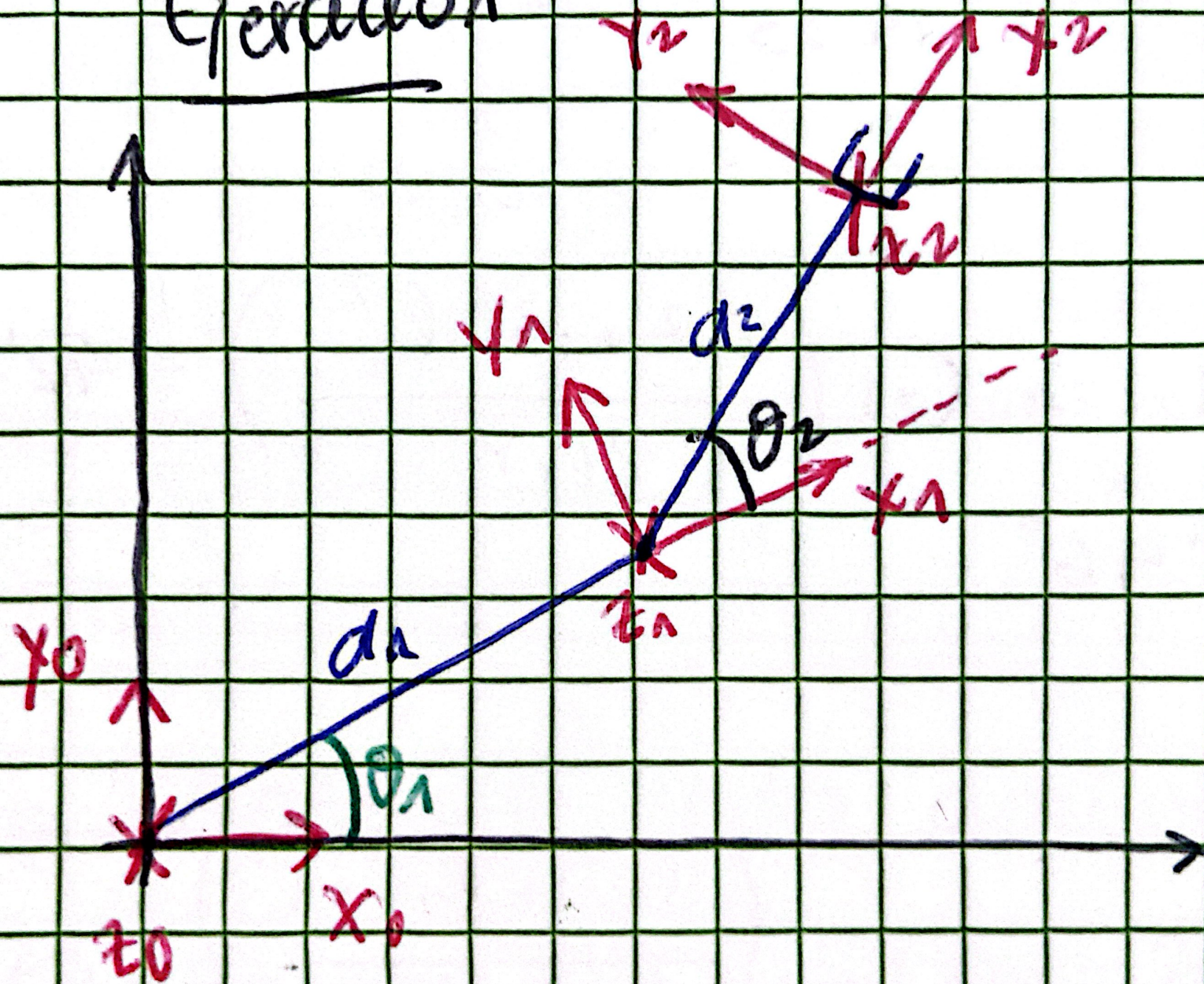


Ejercicio 1



Matriz de Denavit y Hartenberg $\rightarrow$ Ejercicio 1

$\alpha_1 = \alpha_2 = 0 \rightarrow$ Todos los ejes "z" están paralelos.

$d_1 = d_2 = 0 \rightarrow$ los ejes "z" correspondientes se cruzan con los "x" por ello la distancia es igual a 0

Eslabon 1

$$A_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & \sin \theta_1 & a_1 \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 & -\cos \theta_1 & a_1 \sin \theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Eslabon 2

$$A_2 = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & \sin \theta_2 & a_2 \cos \theta_2 \\ \sin \theta_2 & \cos \theta_2 & -\cos \theta_2 & a_2 \sin \theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformación completa $\rightarrow T_0^2 = A_1 A_2$

$$T_0^2 = A_1 A_2 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & a_1 \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 & 0 & a_1 \sin \theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & a_2 \cos \theta_2 \\ 0 & a_2 \sin \theta_2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta_1 \cos \theta_2 & -\sin \theta_1 \sin \theta_2 & -\cos \theta_1 \sin \theta_2 - \sin \theta_1 \cos \theta_2 \\ \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 & \sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\cos(\theta_1 + \theta_2)$ $-\sin(\theta_1 + \theta_2)$

$\sin(\theta_1 + \theta_2)$ $\cos(\theta_1 + \theta_2)$

$$\begin{bmatrix} 0 & (\cos \theta_1 a_2 \cos \theta_2 - \sin \theta_1 a_2 \sin \theta_2) + a_1 \cos \theta_1 \\ 0 & (\sin \theta_1 a_2 \cos \theta_2 + \cos \theta_1 a_2 \sin \theta_2) + a_1 \sin \theta_1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$a_2 \cos(\theta_1 + \theta_2)$ $a_2 \sin(\theta_1 + \theta_2)$

$$T_0^2 = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & a_2 \cos(\theta_1 + \theta_2) + a_1 \cos \theta_1 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & a_2 \sin(\theta_1 + \theta_2) + a_1 \sin \theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$